

02285 AI and MAS, F26

Exercises for week 3, 17/2-26

Exercise 1 (Planes and airports)

Given the action schemas from Figure 1, what are all the applicable concrete instances of $Fly(p, from, to)$ in the state described by

$$At(P1, JFK) \wedge At(P2, SFO) \wedge Plane(P1) \wedge Plane(P2) \wedge \\ Airport(JFK) \wedge Airport(SFO)?$$

Exercise 2 (Simplified Gripper domain)

This exercise refers to the simplified Gripper domain introduced in the lecture on Classical Automated Planning, cf. the slides.

- a) The domain was in the lecture declared using two symmetric actions $MoveAB(x)$ and $MoveBA(x)$. Write the actions schema for a single action $Move$ that is able to replace $MoveAB$ and $MoveBA$. You might have to introduce additional variables and predicates.
- b) Using $Move$ instead of $MoveAB$ and $MoveBA$ is likely to affect also the description of the initial state. Provide the new initial state of the simplified Gripper planning problem with n boxes initially located in location A .
- c) Assume s is a state resulting from the execution of some sequence of $Move$ actions in the initial state. Show that if $In(\alpha, \beta)$ is a literal in s , then α is a constant denoting a box and β is a constant denoting a location. (If this doesn't hold, go back to the first question and revise your action schema).
- d) Does your $Move$ action allow moving a box from a location to itself, e.g. moving box 1 from location A to location A (in one action)? If so, revise your action schema.

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Init(At(C1, SFO) ∧ At(C2, JFK) ∧ At(P1, SFO) ∧ At(P2, JFK)
    ∧ Cargo(C1) ∧ Cargo(C2) ∧ Plane(P1) ∧ Plane(P2)
    ∧ Airport(JFK) ∧ Airport(SFO))
Goal(At(C1, JFK) ∧ At(C2, SFO))
Action(Load(c, p, a),
    PRECOND: At(c, a) ∧ At(p, a) ∧ Cargo(c) ∧ Plane(p) ∧ Airport(a)
    EFFECT: ¬ At(c, a) ∧ In(c, p))
Action(Unload(c, p, a),
    PRECOND: In(c, p) ∧ At(p, a) ∧ Cargo(c) ∧ Plane(p) ∧ Airport(a)
    EFFECT: At(c, a) ∧ ¬ In(c, p))
Action(Fly(p, from, to),
    PRECOND: At(p, from) ∧ Plane(p) ∧ Airport(from) ∧ Airport(to)
    EFFECT: ¬ At(p, from) ∧ At(p, to))

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Figure 1: A PDDL description of an air cargo transportation planning problem.

- e) Draw a diagram of the state space of the simplified Gripper planning problem with $n = 3$. To save space, you can exclude mentioning the rigid predicates in your states. A **rigid predicate** is one that is state invariant like *Box* in the simplified Gripper problem used in the slides. More precisely, an n -ary predicate P is *rigid* if $P(c_1, \dots, c_n)$ for any choice of constants $c_1, \dots, c_n$ will always have the same truth value as in the initial state, independent of the actions applied.
- f) The *branching factor* of a planning problem is the maximum number of actions applicable in any state reachable from the initial state. What is the branching factor of the simplified Gripper problem as a function of n ?
- g) Consider a modified planning problem with three locations, A , B and C . It is assumed that any box can be moved from any location to any other location. Describe the planning problem where all n boxes initially are located in A and where the goal is to get the even-numbered boxes into B and the odd-numbered into C . How much do you need to change: Initial state? Action schema? Goal state?
- h) Modify the planning problem from the previous question further so that a single *Move* can only move a box from A to B , from B to A , from B to C or from C to B . You might again need additional predicates.

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Init( $On(A, Table) \wedge On(B, Table) \wedge On(C, A)$ 
 $\wedge Block(A) \wedge Block(B) \wedge Block(C) \wedge Clear(B) \wedge Clear(C)$ )
Goal( $On(A, B) \wedge On(B, C)$ )
Action(Move( $b, x, y$ ),
  PRECOND:  $On(b, x) \wedge Clear(b) \wedge Clear(y) \wedge Block(b) \wedge Block(y) \wedge$ 
 $(b \neq x) \wedge (b \neq y) \wedge (x \neq y)$ ,
  EFFECT:  $On(b, y) \wedge Clear(x) \wedge \neg On(b, x) \wedge \neg Clear(y)$ )
Action(MoveToTable( $b, x$ ),
  PRECOND:  $On(b, x) \wedge Clear(b) \wedge Block(b) \wedge (b \neq x)$ ,
  EFFECT:  $On(b, Table) \wedge Clear(x) \wedge \neg On(b, x)$ )

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Figure 10.3 A planning problem in the blocks world: building a three-block tower. One solution is the sequence $[MoveToTable(C, A), Move(B, Table, C), Move(A, Table, B)]$.

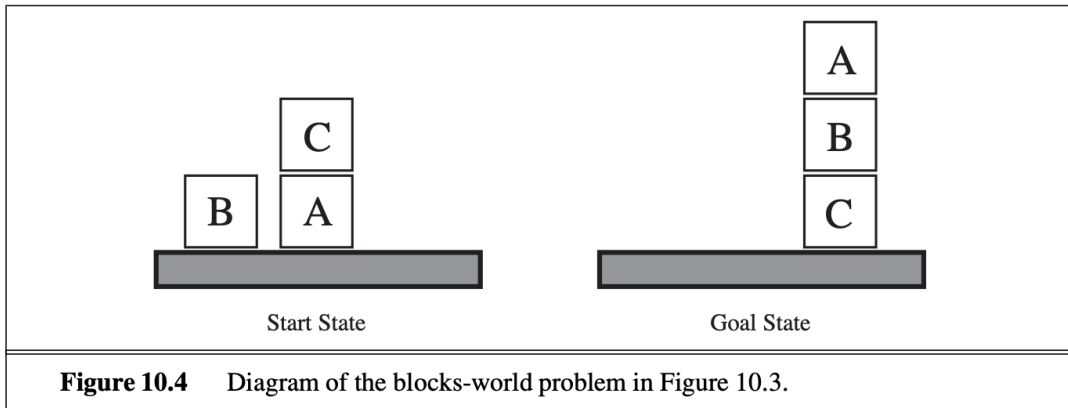


Figure 10.4 Diagram of the blocks-world problem in Figure 10.3.

Figure 2: A simple blocks world problem from R&N, 3ed.

Exercise 3 (Blocks World)

Consider the Blocks World problem P described in PDDL in Figure 2. Let s_0 denote the initial state of this problem, and let g denote the goal.

- The blocks world P problem possesses the so-called **Sussman anomaly**. The problem was considered anomalous because the non-interleaved planners of the early 1970s could not solve it. A non-interleaved planner is a planner that, given two subgoals g_1 and g_2 , produces either a plan for g_1 concatenated with a plan for g_2 , or vice versa. Explain why a non-interleaved planner cannot solve this problem.
- Determine $h^*(s_0)$ (the optimal cost of a solution to the problem).

- c) Determine $h_{gc}(s_0)$ (the goal count heuristic).
- d) Provide an example of a Blocks World problem where $h_{gc}(s_0) = 1$ and $h^*(s_0) \geq 1000$ (that is, a problem where h_{gc} heavily underestimates the optimal cost).
- e) Provide an example of a Blocks World problem where $h_{gc}(s_0) > h^*(s_0)$ (that is, where h_{gc} **overestimates** the optimal cost).
- f) Determine $h_{ip}(s_0)$ (the ignore preconditions and non-goal literal heuristic) for the original Blocks World problem of Figure 10.3.
- g) Provide an example of a Blocks World problem where $h_{ip}(s_0) = 1$ and $h^*(s_0) \geq 1000$ (that is, a problem where h_{ip} heavily underestimates the optimal cost).
- h) Is it possible to create a Blocks World problem where $h_{ip}(s_0) > h^*(s_0)$?
- i) In the initial state of the Blocks World problem of Figure 10.3, the ground atoms of the form $C_1 = C_2$ that hold true are left implicit. Add the relevant atoms of this form to the initial state. In the rest of the exercise, we will assume that these atoms have been added.
- j) The heuristic based on delete-relaxation requires that we have a planning problem where goals and preconditions only contain positive literals. This is not true of the Blocks World problem of Figure 10.3, as an inequality $x \neq y$ is short for $\neg(x = y)$. Rephrase the Blocks World problem so that there are no negative literals in goal and preconditions. In the rest of the exercise, we will work with these rephrased version of the problem.
- k) Determine $h^+(s_0)$ (the delete-relaxation heuristic) for the rephrased Blocks World problem.
- l) Provide an example of a Blocks World problem where $h^+(s_0) < h^*(s_0)$. Explain what it is that make h^+ underestimate the optimal cost in your example.
- m) As defined earlier, a *rigid predicate* is one that is state variant, that is, a predicate P such that no $P(c_1, \dots, c_n)$ can ever get a different truth value than the one it has in the initial state. Find the rigid predicates in the rephrased Blocks World problem. Then show that for any atom $P(c_1, \dots, c_n)$ in any planning problem where P is rigid, we must have either $h_{add}(P(c_1, \dots, c_n), s) = 0$ or $h_{add}(P(c_1, \dots, c_n), s) = \infty$ for all states s , where h_{add} is the additive heuristic.

- n) Determine $h_{add}(s_0)$ (the additive heuristic) for the rephrased Blocks World problem. Explain also the intuitive meaning of all the values $h_{add}(p, s_0)$ you calculate in order to compute $h_{add}(s_0)$. *Hint:* Make use of your answer to the previous question. You can simplify the calculations by not including all the calculations of the atoms using rigid predicates, but simply immediately replace by 0 or ∞ whenever appropriate.
- o) Give an example of a Blocks World problem where $h_{add}(s_0) > h^*(s_0)$.
- p) Determine $h_{max}(s_0)$ (the max heuristic) for the rephrased Blocks World problem. Explain also the intuitive meaning of all the values $h_{max}(p, s_0)$ you calculate in order to compute $h_{max}(s_0)$.
- q) Is it possible to create a Blocks World problem where $h_{max}(s_0) > h^*(s_0)$?
- r) Find the admissible heuristics among the ones you have considered above (except h^*). Taking the maximum value of the value returned by all of these still gives an admissible heuristic (cf. the slides). What is this maximum value for the (rephrased) Blocks World problem considered in this exercise?
- s) Compute $h_{FF}(s_0)$. When constructing the planning graph, you're allowed to omit states and actions that are not necessary to compute the value of $h_{FF}(s_0)$. Is any relaxed solution extracted from the planning graph a real solution to the problem?
- t) *Optional:* Draw the (relevant part of the) directed hypergraph for calculating $h_{add}(s_0)$, cf. the slides from today and Section 2.7 of Geffner & Bonet. Look up directed hypergraph on Google first, if you don't know what they or how to draw them. To make the graph smaller, you can omit all nodes containing rigid predicates, as well as omit any action a in which one of the preconditions is false in s_0 and uses a rigid predicate.